Lower bound for twisted k-jet sections through an approximate Zariski decomposition

Campana, Frederic; Darondeau, Lionel; Demailly, Jean-Pierre; Rousseau, Erwan

doi:10.5802/ahl.197; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/ahl-197-on-the-existence-of-logarithmic-and-orbifold-jet-2024/pinned-original-source.tex (On the existence of logarithmic and orbifold jet differentials); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Published printed number is Corollary 5.3. The bracketed mention [Fuj94] in the proof is a literal source text token, not a \cite command: the span carries zero \cite and its only \ref is the corollary's own label (self-citation at the end of the proof, resolves internally). All displays inside the span are unnumbered; no labelled equations.

Editorial note. The label coro5.3 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own coro environment in the article source.

Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, no bibliography in the counted span (the span carries no \cite commands); references leaving the span are declared as navigation stubs carrying the article's printed numbers.

Editorial scope. the counted statement and its own proof environment only

Corollary 5.3. *If F is an arbitrary $\mathbb{Q}$-line bundle over X, one has*

$$h^0\Big(X_k(V), \mathcal{O}_{X_k(V)}(m) \otimes \pi_k^* \mathcal{O}\Big(-\tfrac{m}{kr}\Big(1+\tfrac{1}{2}+\cdots+\tfrac{1}{k}\Big)F\Big)\Big) \geqslant \tfrac{m^{n+kr-1}}{(n+kr-1)!}\tfrac{(\log k)^n}{n!\,k!^r}\left(\mathrm{Vol}(K_V \otimes F^{-1}) - O\left((\log k)^{-1}\right)\right) - o\left(m^{n+kr-1}\right),$$

k-jet differentials of degree m twisted by the appropriate power of F if $K_V \otimes F^{-1}$ is big.

Proof. The volume is computed here as usual, i.e. after performing a suitable modification $\mu : \widetilde{X} \to X$ which converts K_V into an invertible sheaf. There is of course nothing to prove if $K_V \otimes F^{-1}$ is not big, so we can assume $\mathrm{Vol}(K_V \otimes$

$F^{-1}) > 0$. Let us fix smooth hermitian metrics h_0 on T_X and h_F on F. They induce a metric $\mu^*(\det h_0^{-1} \otimes h_F^{-1})$ on $\mu^*(K_V \otimes F^{-1})$ which, by our definition of K_V, is a smooth metric. By the result of Fujita [Fuj94] on approximate Zariski decomposition, for every $\delta > 0$, one can find a modification $\mu_\delta : \widetilde{X}_\delta \to X$ dominating μ such that

$$\mu_\delta^* \left(K_V \otimes F^{-1}\right) = \mathcal{O}_{\widetilde{X}_\delta}(A + E)$$

where A and E are $\mathbb{Q}$-divisors, A ample and E effective, with

$$\mathrm{Vol}(A) = A^n \geqslant \mathrm{Vol}\left(K_V \otimes F^{-1}\right) - \delta.$$

If we take a smooth metric h_A with positive definite curvature form Θ_{A,h_A}, then we get a singular hermitian metric $h_A h_E$ on $\mu_\delta^*(K_V \otimes F)$ with poles along E, i.e. the quotient $h_A h_E / \mu^*(\det h_0^{-1} \otimes h_F)$ is of the form $e^{-\varphi}$ where φ is quasi-psh with log poles $\log|\sigma_E|^2 \pmod{C^\infty(\widetilde{X}_\delta)}$ precisely given by the divisor E. We then only need to take the singular metric h on T_X defined by

$$h = h_0 e^{\frac{1}{r}(\mu_\delta)^*\varphi}$$

(the choice of the factor $\frac{1}{r}$ is there to correct adequately the metric on $\det V$). By construction h induces an admissible metric on V and the resulting curvature current $\eta_F = \Theta_{K_V, \det h^*} - \Theta_{F,h_F}$ is such that

$$\mu_\delta^* \eta_F = \Theta_{A,h_A} + [E], \qquad [E] = \text{current of integration on } E.$$

Then the 0-index Morse integral in the complement of the poles is given by

$$\int_{X(\eta,0)\setminus S} \eta_F^n = \int_{\widetilde{X}_\delta} \Theta_{A,h_A}^n = A^n \geq \mathrm{Vol}\left(K_V \otimes F^{-1}\right) - \delta$$

and Corollary 5.3 follows from the fact that δ can be taken arbitrary small. $\square$